PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

Smart Complaint Management System

Version 1.1

Prepared: 24-10-2026

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Revision History

Name	Date	Reason For Changes	Version
Project Team	06-10-2026	Initial SRS for mini-project	1.0
Project Team	24-10-2026	Issue #69: rebuilt the requirement traceability matrix (Appendix B) with issue references, implementation evidence and test functions; added the legacy test case crosswalk (Appendix C), requirement notes (3.7) and implementation notes under 2.4 and 3.1.2. Requirement IDs and wording are unchanged.	1.1

1. Introduction

1.1 Purpose

This SRS specifies the requirements for the Smart Complaint Management System, a web-based platform for lodging, tracking and resolving complaints. It defines scope, users, interfaces, functional/non-functional requirements, security requirements and traceability.

1.2 Intended Audience

Development team, project owner, testers, faculty evaluators and maintainers.

1.3 Product Scope

The system centralizes complaint submission and administration. A registered user can create a complaint with title, description and category, view submitted complaints and track status. An administrator can view/filter complaints, update status and record resolution information.

1.4 References

- SE Mini-Project Deliverables Part-1, PES University, 2026.
- Supplied PES University SRS template.
- Supplied PES University SAD template.
- Supplied PES University Test Plan template.

1.5 Overview

Section 2 describes the product and users. Section 3 specifies interfaces, functional/non-functional requirements, security, quality attributes, business rules and domain requirements. Appendices provide a glossary and requirement traceability matrix.

1.6 Version 1.1 Baseline and Evidence

Version 1.1 does not change any requirement. The requirement text and IDs in section 3 are those of version 1.0. What changed is listed in the revision history.

The implementation evidence in this version describes the `main` branch at commit `2cc96e3` (local reference). Open pull requests #83, #84, #85, #86 and #87 are pending work and are not treated as implemented.

Evidence is labelled **code inspection** (read from the source on `main`). No tests were run and no routes were executed for this version, so no test result has been recorded: every result in Appendix B is "Not run". The Test Plan (version 1.0) is unchanged and is cited by its original test case numbers. No statement is made about any hosted environment.

2. Overall Description

2.1 Product Perspective

The proposed system is a new self-contained web application. The logical flow is from the User to the Web UI, then to the Application/API, and then to the Database, with an Administrator using the same application through role-based access.

2.2 Product Functions

- Registration/login/logout
- Complaint creation and validation
- View and track own complaints
- Administrator login and role-based access
- Administrator complaint listing/filtering
- Status updates and resolution remarks
- Persistent user and complaint records
- Clear error feedback

2.3 User Classes and Characteristics

User Class	Characteristics	Main Functions
Registered User	Basic web literacy; authenticated account.	Register/login, submit complaints, view own complaints, track status.
Administrator	Authorized staff with elevated privileges.	View/manage complaints, update status, record resolution.

2.4 Operating Environment

Modern desktop/mobile browser, backend application server and persistent database. The exact backend framework and database should be finalized by the team.

Note (version 1.1): the implementation uses Python with Flask, Flask-SQLAlchemy and SQLite; see SAD version 1.1, section 2.7.

2.5 Design and Implementation Constraints

- Only authenticated users may submit complaints.
- Administrator functions are role restricted.
- Status changes follow the supported workflow.
- Secrets must not be hard-coded or committed.
- Implementation must remain practical for the mini-project environment.

2.6 Assumptions and Dependencies

- Users have browser/network access.
- Database is available when the application runs.
- At least one administrator account is provisioned.
- Deployment and third-party services may be finalized during implementation.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interface

- Login/registration clearly shows validation errors.
- Complaint form has labelled mandatory fields.
- User dashboard displays complaint status/details.
- Admin dashboard provides listing, filtering and management controls.

3.1.2 Software Interfaces

- Frontend communicates with backend through HTTP/HTTPS APIs.
- Backend communicates with the selected database.
- Authentication state uses the selected secure session/token mechanism.

Note (version 1.1): the application renders HTML pages on the server and accepts form posts; it has no JSON API. Authentication state is a signed Flask session cookie. See SAD version 1.1, sections 2.6 and 3.4.

3.1.3 Communications Interfaces

- HTTP/HTTPS for client-server communication.
- HTTPS should be used in deployment.

3.2 Functional Requirements

ID	Feature	Requirement	Priority
FR-01	User Registration	Allow a new user to create an account using required information.	High
FR-02	User Authentication	Authenticate registered users before protected functions are available.	High
FR-03	Complaint Submission	Allow an authenticated user to submit a complaint with title, description and category.	High
FR-04	Complaint Validation	Reject submissions when mandatory fields are missing or invalid.	High
FR-05	View Own Complaints	Allow a user to view complaints submitted by that user.	High
FR-06	Track Complaint	Display current status and details of a user's complaint.	High
FR-07	Admin Access Control	Restrict administrative functions to authenticated administrator accounts.	High
FR-08	Admin Complaint Management	Allow administrators to view and filter complaint records.	High
FR-09	Status Update	Allow administrators to update complaint status using the supported workflow.	High
FR-10	Resolution Details	Allow administrators to record resolution remarks.	Medium
FR-11	Data Persistence	Persist users and complaints so records remain after a session ends.	High
FR-12	Error Feedback	Provide clear error messages for failed operations or invalid input.	Medium

3.3 Non-Functional Requirements

ID	Attribute	Measurable Requirement
NFR-01	Performance	For normal load, 95% of user-facing API requests should respond within 2 seconds.
NFR-02	Availability	Deployed system should achieve at least 99% availability during evaluation, excluding scheduled maintenance.
NFR-03	Security	Passwords shall use secure one-way hashing and never be stored in plaintext.
NFR-04	Authorization	Protected operations shall verify the user's role before privileged actions.
NFR-05	Confidentiality	Credentials and complaint data shall use HTTPS in deployment.
NFR-06	Usability	A first-time user should locate complaint submission without administrator assistance.
NFR-07	Reliability	A failed database/API operation shall not create a misleading successful complaint state.
NFR-08	Maintainability	Frontend, backend and data-access responsibilities shall be modular.
NFR-09	Scalability	Design should permit additional users/categories/admins without redesigning the workflow.
NFR-10	Testability	Functional requirements shall be testable using repeatable test cases.

3.4 Security Requirements

ID	Requirement
SEC-01	Authenticate users before protected access.
SEC-02	Enforce role-based authorization for administrator operations.
SEC-03	Do not store or return passwords in plaintext.
SEC-04	Validate and sanitize user-supplied complaint fields.
SEC-05	Do not expose sensitive account information in authentication errors.
SEC-06	Keep database credentials and token keys outside committed source code.

3.5 Software Quality Attributes

- Correctness: ownership/status data remains consistent.
- Usability: forms and statuses are understandable.
- Maintainability: modules have clear responsibilities.
- Reliability: failures do not silently lose data.
- Security: authentication, authorization and secure credential handling are mandatory.

3.6 Business and Domain Rules

- A complaint belongs to the authenticated user who submitted it.
- Only an administrator can change administrative status.
- Resolved complaints should contain resolution information.
- Users only see their own complaints unless an explicit public/administrative view exists.

3.7 Requirement Notes (version 1.1)

Observations where the wording of a requirement does not match the implementation on `main`. The requirement text above is unchanged and no requirement was remapped. All observations are from code inspection.

Item	Observation	Related
FR-06	The complaint list shows the status and resolution, but there is no complaint detail page, so "details" are limited to what the list shows.	#26, #51; PR #83 pending
FR-09, 2.5	"Supported workflow": any of the three statuses can be set from any other. The only rules are that the status must be one of the three values and that Resolved needs remarks. There is no transition graph.	#31, #43
FR-11	Records persist in a local SQLite file. Persistence of a hosted deployment is not verified by the repository.	#2, #73; PR #85 pending
NFR-01	The application has no API. The wording refers to page requests, and no response-time measurement exists in the repository.	#23
NFR-05	The repository has only a `SESSION_COOKIE_SECURE` switch. Issue #24 is titled NFR-05 but covers the secret key and debug mode, which corresponds to SEC-06.	#24, #39
SEC-04	Complaint fields are trimmed and checked for non-empty; output is autoescaped by Jinja2. There are no length or content rules.	#20
SEC-05	The login message is generic, but registration reports that an email is already registered.	#3
3.1.1	There is no separate user dashboard page on `main`; status is shown in the complaint list.	#27, #66; PR #87 pending
FR-09, FR-10	The admin complaint page cannot be rendered because its template builds a link with an endpoint name that does not exist (SAD version 1.1, section 3.8, item 8). Not executed.	#41

Appendix A: Glossary

Term	Meaning
Complaint	User-submitted issue requiring attention.
Administrator	Authorized user who manages complaints.
Status	Current processing stage such as Pending, In Progress or Resolved.
RBAC	Role-Based Access Control.
API	Application Programming Interface.
SRS	Software Requirements Specification.

Appendix B: Requirement Traceability Matrix

One row per requirement of section 3. Issue references come from the titles on the public issue list; a closed issue is not treated as proof that the feature exists. "Implementation evidence" is code inspection of `main` at commit `2cc96e3`. Test functions are in `tests/test_app.py` on `main`. **Result** "Not run" means that no test was executed for this document and no outcome is claimed; it does not say that a test exists. The Test Plan numbers (TC-xx) are mapped in Appendix C.

Req ID	Requirement summary	Issue ref	Implementation evidence	Test function	Result	Notes
FR-01	User registration	#3; #68 (open)	`auth.register`	`test_tc01_registration`; `test_tc02_duplicate_registration`	Not run	Fields are only checked for non-empty; duplicate check is case-sensitive. #68: PR #86 pending.
FR-02	User authentication	#4, #5, #10; #56	`auth.login`, `auth.logout`	`test_tc03_valid_login`; `test_tc04_invalid_login`	Not run	Logout has no test. #56 (missing form fields) not evidenced on `main`.
FR-03	Complaint submission	#6	`complaints.complaint`	`test_tc05_complaint_submission`	Not run	
FR-04	Complaint validation	#20	`complaints.complaint` (non-empty check)	`test_tc06_empty_complaint_rejected`	Not run	Test asserts only that the word "required" appears.
FR-05	View own complaints	#22, #7	`complaints.complaints`	`test_tc07_user_complaints`	Not run	Test asserts that the submitted title appears.
FR-06	Track complaint	#7, #33; #26, #51	`complaints.complaints`; `Complaint` status, resolution and timestamp fields	None identified	Not run	No detail page on `main` (PR #83 pending). #26 is closed but not evidenced on `main`.
FR-07	Admin access control	#9, #10, #15, #16	`admin.admin_only`; 403 handler and template	`test_tc08_admin_access`; `test_tc09_normal_user_denied_admin`	Not run	
FR-08	Admin complaint management	#11, #12, #32; #42	`admin.admin_dashboard`	`test_tc10_admin_can_view_dashboard`	Not run	Filtering has no test. #42 (category filter) is open.
FR-09	Status update	#13, #31; #41, #43	`admin.admin_complaint`	None identified	Not run	No transition graph. The page cannot be rendered (see 3.7). #31 is closed but the transition rule is not evidenced.
FR-10	Resolution details	#14; #45	`admin.admin_complaint`	None identified	Not run	Resolved requires remarks. No history model; #45 is open.
FR-11	Data persistence	#2; #73	`models.py`; SQLite setting in `app.py`	None identified	Not run	Local SQLite file only. #73 open; PR #85 pending.
FR-12	Error feedback	#21; #52	Messages in `auth.py`, `complaints.py`, `admin.py`; error templates	`test_tc02_duplicate_registration`; `test_tc04_invalid_login`; `test_tc06_empty_complaint_rejected`	Not run	Tests check message text only. #52 (success messages) is open.

Req ID	Requirement summary	Issue ref	Implementation evidence	Test function	Result	Notes
NFR-01	Performance	#23	None	None identified	Not run	No measurement in the repository. Plan TC-11 has no test.
NFR-02	Availability	#39	None	None identified	Not run	No deployment evidence in the repository.
NFR-03	Password hashing	#5, #23	Werkzeug hashing in `auth.py`	None identified	Not run	Plan TC-12 has no test.
NFR-04	Authorization	#15	`admin.admin_only`	`test_tc09_normal_user_denied_admin`	Not run	
NFR-05	Confidentiality (HTTPS)	#24, #39	`SESSION_COOKIE_SECURE` switch in `app.py`	None identified	Not run	Questionable mapping: #24 is titled NFR-05 but covers the secret key and debug mode (see SEC-06). HTTPS is not evidenced.
NFR-06	Usability	#27, #36; #66	None on `main`	None identified	Not run	Navigation and dashboard are pending (#66, PR #87). #27 and #36 are closed but not evidenced.
NFR-07	Reliability	None identified	The 500 handler rolls the database session back	None identified	Not run	Code only.
NFR-08	Maintainability	#37	Blueprints `auth`, `complaints`, `admin`; `models.py`; `extensions.py`	None identified	Not run	Structural; no test.
NFR-09	Scalability	None identified	None	None identified	Not run	Not assessed.
NFR-10	Testability	#34; #59	`tests/test_app.py` (ten tests)	`test_tc01_registration` to `test_tc10_admin_can_view_dashboard`	Not run	Coverage tooling (#59) not evidenced.
SEC-01	Authenticate before protected access	#5	Session checks in `complaints.py`; `admin.admin_only`	None identified	Not run	
SEC-02	Role-based authorization	#15	`admin.admin_only`	`test_tc09_normal_user_denied_admin`	Not run	
SEC-03	No plaintext passwords	#5	Werkzeug hashing in `auth.py`	None identified	Not run	

Req ID	Requirement summary	Issue ref	Implementation evidence	Test function	Result	Notes
SEC-04	Validate complaint fields	#20	`complaints.complaint` (trim, non-empty); Jinja2 autoescaping	`test_tc06_empty_complaint_rejected`	Not run	No length or content rules.
SEC-05	No sensitive data in auth errors	None identified	Generic login message in `auth.login`	`test_tc04_invalid_login`	Not run	Registration says "Email already registered".
SEC-06	Secrets outside source	#24	`SECRET_KEY` read from the environment in `app.py`	None identified	Not run	No database credentials exist (SQLite).

Appendix C: Legacy Test Plan Crosswalk

The Test Plan (version 1.0) is unchanged. Its test case numbers (TC-01 to TC-12) differ from the numbers in the test function names, so the cases are matched here by scenario. The Result for every row is "Not run".

Plan ID	Plan scenario	Plan requirement	Test function on `main`	Note	Result
TC-01	Valid registration	FR-01	`test_tc01_registration`	Same number	Not run
TC-02	Valid login	FR-02	`test_tc03_valid_login`	Function number differs	Not run
TC-03	Invalid login	FR-02	`test_tc04_invalid_login`	Function number differs	Not run
TC-04	Submit complaint	FR-03	`test_tc05_complaint_submission`	Function number differs	Not run
TC-05	Missing fields	FR-04	`test_tc06_empty_complaint_rejected`	Function number differs	Not run
TC-06	View own complaints	FR-05/06	`test_tc07_user_complaints`	Function number differs; no detail page	Not run
TC-07	Unauthorized admin access	FR-07/SEC-02	`test_tc09_normal_user_denied_admin`	Function number differs	Not run
TC-08	Admin opens and filters complaint list	FR-08	`test_tc10_admin_can_view_dashboard`	Partial: the test opens the dashboard but does not filter	Not run
TC-09	Status update	FR-09	None identified	No test on `main`	Not run
TC-10	Resolution	FR-10	None identified	No test on `main`	Not run
TC-11	Response time	NFR-01	None identified	No test on `main`	Not run
TC-12	Credential/secret protection	NFR-03	None identified	No test on `main`	Not run
(none)	Duplicate registration	FR-01	`test_tc02_duplicate_registration`	Not in the Test Plan	Not run
(none)	Admin login redirect	FR-07	`test_tc08_admin_access`	Not in the Test Plan	Not run